DC Motor

Permanent-Magnet Brushed DC Motor

Part Number: Assumed Equivalent — Mabuchi RS-385SH-2270 (12 V)

Identification note: *The uploaded Proteus schematic uses Proteus's generic simulation motor model ("MOTOR-DC", model file DCMOTOR.MDF) with a 12 V supply rail and a zero-load speed parameter of 1000 RPM configured in the simulation. This is a simulation placeholder, not a real manufacturer part number, since Proteus motor models do not reference actual commercial part numbers. The closest commonly-used real-world equivalent for a 12 V brushed DC motor of this class in student/hobbyist H-Bridge projects is the* ***Mabuchi RS-385SH-2270****, and it is used here as the assumed equivalent component, with all electrical data drawn from its official manufacturer specification sheet.*

1. Component Name

12 V Permanent-Magnet Brushed DC Motor — Assumed equivalent part: Mabuchi RS-385SH-2270 (carbon-brush type). Marked as an assumed/closest-equivalent component because the schematic defines only a generic simulation motor block, not a specific manufacturer part number.

2. Introduction

A brushed DC motor is an electromechanical device that converts electrical energy into rotational mechanical energy using a permanent-magnet stator, a wound rotor (armature), and a mechanical commutator with brushes that periodically reverse current in the rotor windings to sustain continuous rotation. Brushed DC motors are popular in low-cost robotics, hobby, and educational projects because their speed and direction can be controlled simply by adjusting the polarity and average voltage applied across their two terminals.

3. Manufacturer

Manufacturer	Mabuchi Motor Co., Ltd.
Official Datasheet / Product	RS-385SH-2270 Carbon-Brush DC Motor
Datasheet Reference	Mabuchi Motor product datasheet, RS-385SH series
Status	Assumed / closest commonly-used equivalent (see identification note above)

4. Functional Description

Working Principle

When a DC voltage is applied across the motor terminals, current flows through the armature winding situated inside the magnetic field of the stator's permanent magnets. This current-carrying conductor experiences a force (per the motor effect / Lorentz force) that produces torque and causes the rotor to spin. The mechanical commutator continuously switches the current direction in the rotor coils as the shaft rotates so that torque is generated in a consistent direction, sustaining continuous rotation.

Purpose / Why it is used in this H-Bridge DC Motor Driver with PWM project

The DC motor is the actuator (load) of the entire project — it is the component whose speed and direction the H-Bridge and PWM signal are designed to control. Reversing the polarity applied by the H-Bridge changes the motor's direction of rotation, while varying the PWM duty cycle applied to the switching MOSFETs varies the average voltage seen by the motor and therefore its speed.

5. Key Features

Feature	Specification
Nominal Operating Voltage	12 V DC
Operating Voltage Range	6 V - 24 V DC
No-Load Speed	Approx. 9900 - 10000 RPM
No-Load Current	0.20 A
Rated (Continuous) Output	0.9 W - 35 W (approx., model dependent)
Commutation Type	Mechanical brush / carbon-brush commutator
Package / Frame Size	Approx. dia. 27.7 mm x 37.8 mm (385-frame)
Weight	Approx. 70 g

6. Pin / Terminal Configuration

Terminal	Name	Function
1	Terminal A (+)	Positive motor terminal in one rotation direction; reversing polarity relative to Terminal B reverses rotation direction
2	Terminal B (-)	Second motor terminal; current return path / opposite polarity terminal

7. Electrical Specifications

Parameter	Symbol	Value	Conditions
Rated Voltage	V	12 V DC	Nominal
No-Load Speed	N0	~9900-10000 RPM	At rated voltage
No-Load Current	I0	0.20 A	At rated voltage
Stall Torque	Ts	43.2 mN.m (440 gf.cm)	At rated voltage
Stall Current	Is	4.00 A	At rated voltage
Operating Voltage Range	-	6 V - 24 V DC	Manufacturer rated range

8. Typical Applications

- Robotics wheel/drive actuation
- H-Bridge / PWM-based motor speed and direction control projects
- Small home appliances (hair dryers, mixers)
- Toy and hobbyist mechanical actuation
- Printers and office equipment paper-feed mechanisms
- Automotive auxiliary systems (window lifts, small pumps)
- Conveyor belts and small automation equipment
- Educational demonstrations of motor control and PWM concepts

9. Advantages

- Simple two-terminal control — speed and direction fully controlled by applied voltage polarity/magnitude

- Low cost and widely available
- High starting torque relative to size
- Compatible with simple analog or PWM-based drive electronics such as an H-Bridge
- Wide operating voltage range provides design flexibility

10. Limitations

- Mechanical brushes wear over time, limiting operational lifetime and requiring maintenance
- Brush arcing generates electrical noise (EMI) that can affect nearby circuitry
- Lower efficiency compared to brushless DC motors
- Stall current can be significantly higher than rated current, requiring the driving MOSFETs and power supply to tolerate high inrush/stall conditions
- Speed is load-dependent and can vary under changing mechanical load without closed-loop feedback

11. Role in This Project

The DC motor is the final load driven by the H-Bridge. Two diagonal MOSFETs in the bridge are switched on to apply voltage across the motor terminals in a given polarity, spinning the motor in one direction; switching the opposite diagonal pair reverses the polarity and hence the direction of rotation. Simultaneously, the PWM signal generated by the STM32F103C6 microcontroller rapidly switches one leg of the bridge on and off, controlling the average voltage — and therefore the rotational speed — delivered to the motor.

12. Precautions

- Always include a flyback/free-wheeling diode path (or rely on MOSFET body diodes) across the motor to protect switching devices from inductive voltage spikes
- Do not exceed the motor's rated voltage or stall current for extended periods, as this can overheat the windings
- Ensure adequate mechanical mounting to avoid vibration-induced wear on the brushes and bearings
- Account for high inrush/stall current when sizing the MOSFETs, power supply, and PCB traces
- Keep motor brush noise away from sensitive analog/digital circuitry through decoupling and, where possible, physical separation

13. Official Reference

Official Manufacturer	Mabuchi Motor Co., Ltd.
Official Datasheet Title	RS-385SH-2270 Carbon-Brush DC Motor Specification Sheet
Datasheet Version	Manufacturer product datasheet (RS-385SH series)
Component Status	Assumed closest commonly-used equivalent (see note on Page 1)